

Particle Methods

Many success stories in high-dimensional inference!

Sampling & MCMC:

Langevin algorithm (unadjusted, metropolis adjusted, underdamped, ...)
Forward-Backward diffusion, DDPM (Song, Ermon 2025, Ho et al. 2020)
SVGD (Liu, Wang 2016, Liu 2017)

Mixture Modeling:

Variational Inference (Lambert et al. 2022)
Approximate NPMLE (Yan et al. 2023)

Filtering, Numerical PDE:

Ensemble/Particle Kalman Filtering, Sequential MCMC (Doucet et al. 2001)
Smoothed Particle Hydrodynamics (J. Monaghan, 2005)
Measure Quantization (Xu et al, 2022, Belhadji 2025)

Computer Graphics...?

The “Gaussian Splatting” Revolution

Novel View Synthesis: given 2D views + camera positions...

